

Background

- Postinflammatory hyperpigmentation (PIH) is a skin condition that causes discoloration due to excess melanin production after a skin injury, inflammation, or trauma.

- Macrophage Migration Inhibitory Factor (MIF) is involved in regulating the inflammatory response and immune system. MIF plays a crucial role in preventing macrophages from excessively migrating. ISO-1 is a MIF inhibitor that disrupts MIF's proinflammatory activity.

- The B16-F10 cell line is used as a model for hyperpigmentation and other pigmentation conditions. Research performed by Culp et al. (2007) used B16-F10 cells to determine MIF's effects on melanoma, which is similar to PIH as it increases melanin production.

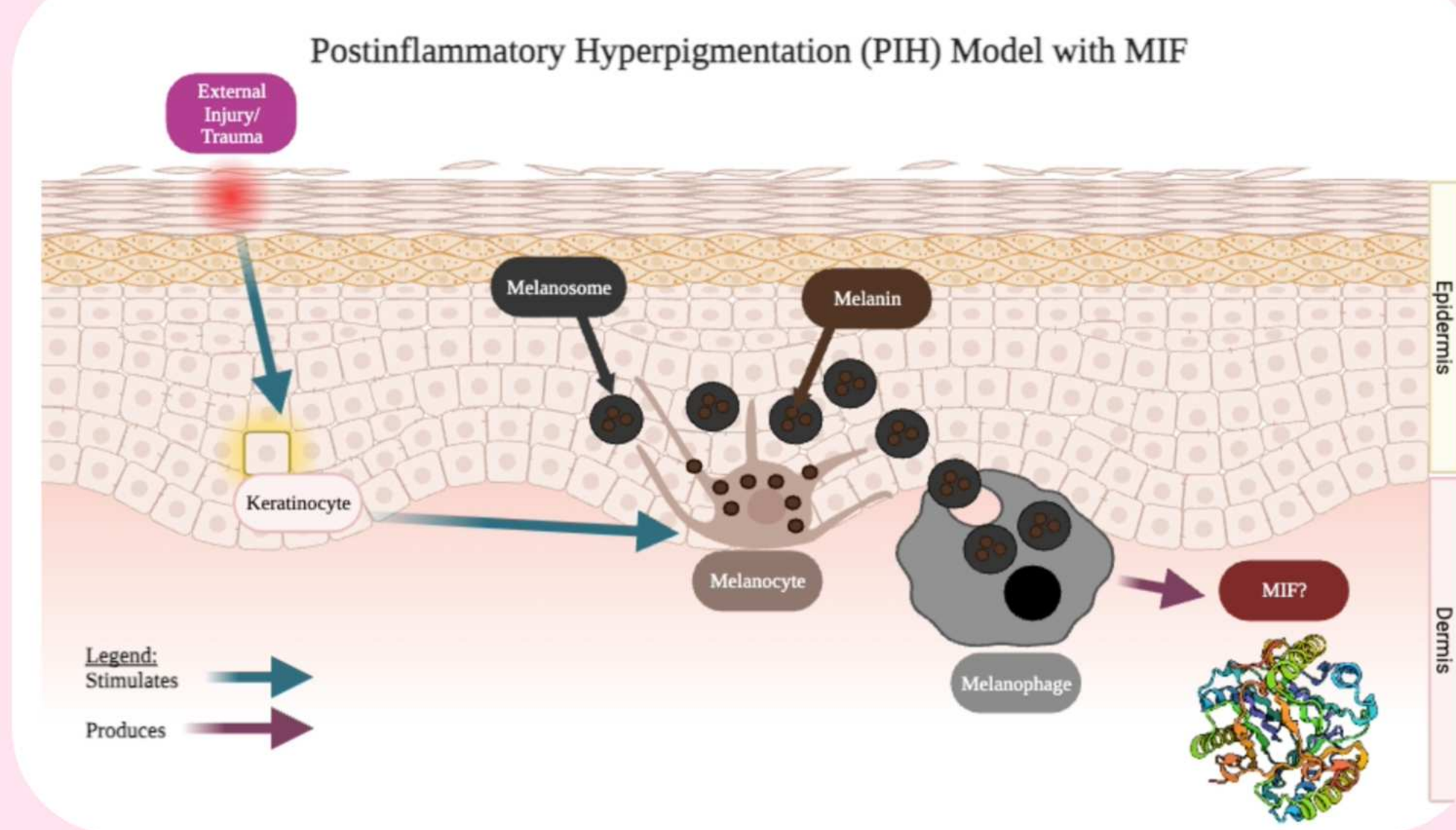


Figure 1. Model of Postinflammatory Hyperpigmentation
Image Created by Student Using BioRender

Problem & Hypothesis

Post-inflammatory hyperpigmentation (PIH) is a prevalent skin condition that affects people of all ages and ethnicities. PIH can result from various chronic skin inflammation diseases such as lichen planus, lupus erythematosus, and atopic dermatitis. Macrophage migration inhibitory factor (MIF) is a protein that is released by macrophages during an inflammatory response and is also part of the melanin biosynthesis pathway. By discovering alternative ways to treat PIH, dermatologists can offer new treatments while researchers can explore new possibilities for treatment. If the amount of ISO-1, an MIF inhibitor, is added to a post-inflammatory hyperpigmentation model, then the melanin production will decrease because MIF has been shown to increase melanin biosynthesis.

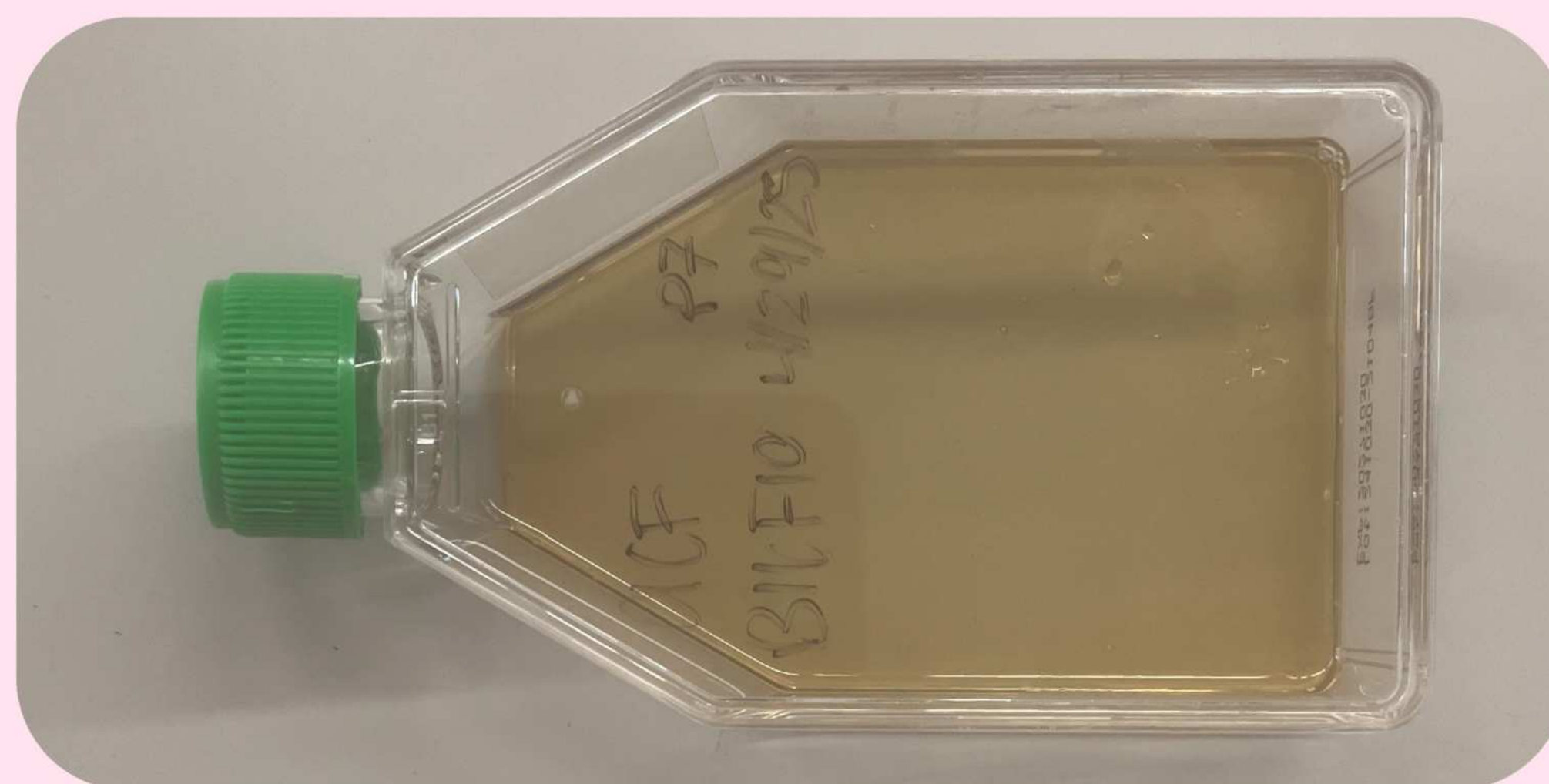


Figure 4. Flask of Cells Producing Melanin
Picture Taken by Student

Determining if MIF Contributes to Postinflammatory Hyperpigmentation Using B16-F10 Cells

TMED

Method

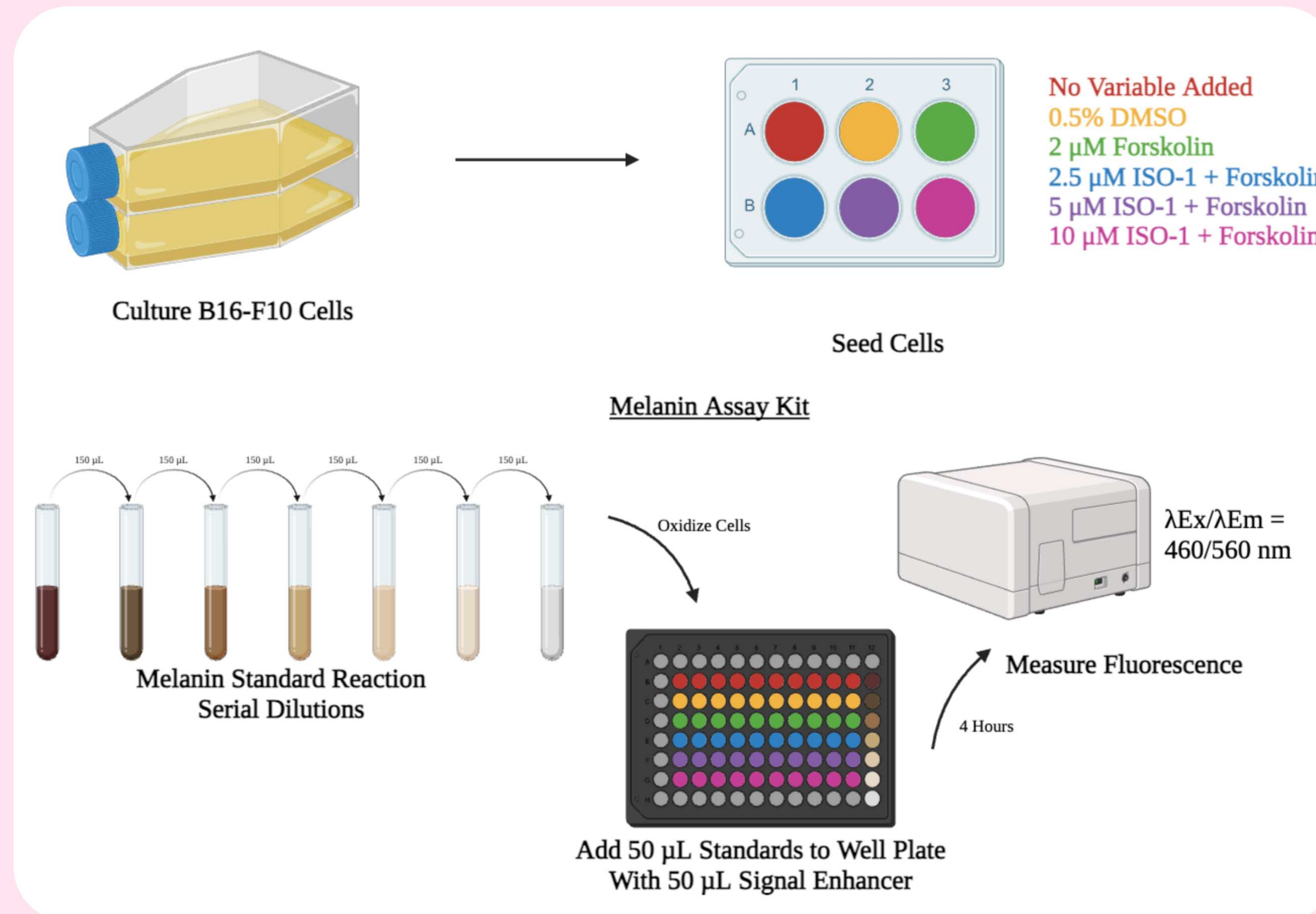
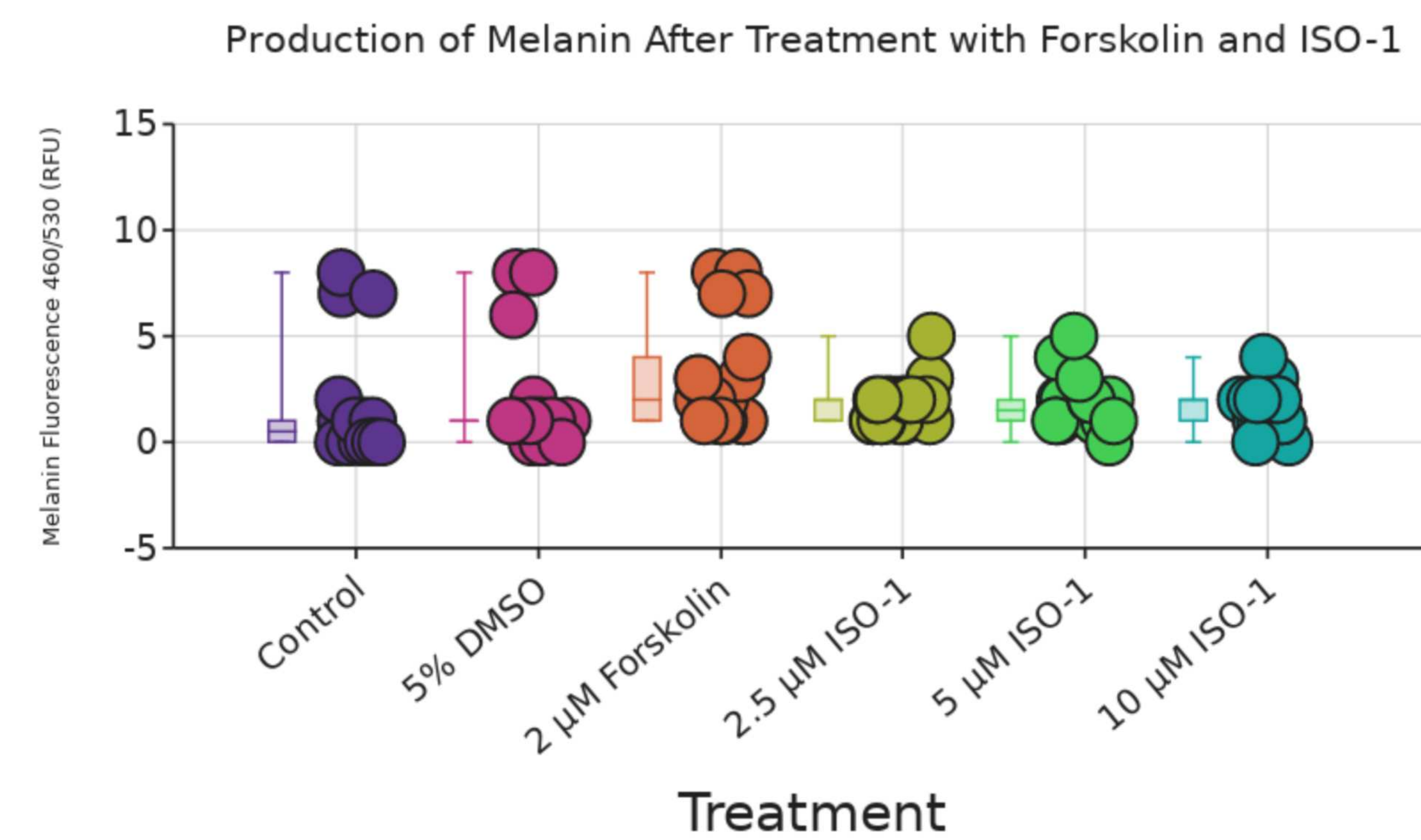


Figure 2. Method
Image Created by Student Using BioRender

Results



Note: Box and whiskers plot shows the median value (line), interquartile range (box), and the extent of the data (whiskers above and below at min / max data points).

All ISO-1 concentrations contain 2 μM Forskolin

Figure 3. Graph Demonstrating the Effects of ISO-1 on a Forskolin Induced Hyperpigmentation Model
Graph Created by Student Using DataClassroom

Discussion

A Kruskal-Wallis test was performed on the data. The p-value was 0.01, meaning that there was a very strong confidence that the groups were different. After a Dunn's test (with Bonferroni correction) was run, the only groups with a statistically significant difference were the Control - 2μM Forskolin, Control - 2.5μM ISO-1, and 5% DMSO - 2μM Forskolin. The Control - 2μM Forskolin had a p-value of <0.01, which showed an extremely strong confidence that the groups are different. Additionally, the 5% DMSO - 2μM Forskolin had a p-value of 0.11, meaning that there was some indication that the groups might have been different. These two differences demonstrated that the model worked because Forskolin increased the melanin concentration between the control groups. On the other hand, the rest of the group's p-values ranged from 0.48 to 1.00, which showed that there was no statistically significant difference between the groups. This shows that ISO-1 was not an effective treatment in reducing melanin production. The only somewhat statistically significant difference with ISO-1 was Control - 2.5μM ISO-1 with a p-value of 0.15, which showed some indication that the groups might have been different. **Overall, Forskolin was an effective stimulus to increase melanin production, but ISO-1 was not an effective treatment in melanin reduction.**

Even though the p-value demonstrated that some groups were different, the assay did not work properly, meaning the data is inconclusive. The melanin within the cells was not fully transferred into the supernatant, so the fluorescent readings weren't accurate. The lack of data was additionally due to the cells not producing melanin within the cells, so previous trials did not work. The cells needed to be incubated for multiple days until there was visually melanin in the cells and supernatant. The melanin assay kit's assay buffer also didn't properly lyse the cells, so the melanin stayed in the cells instead of the supernatant.

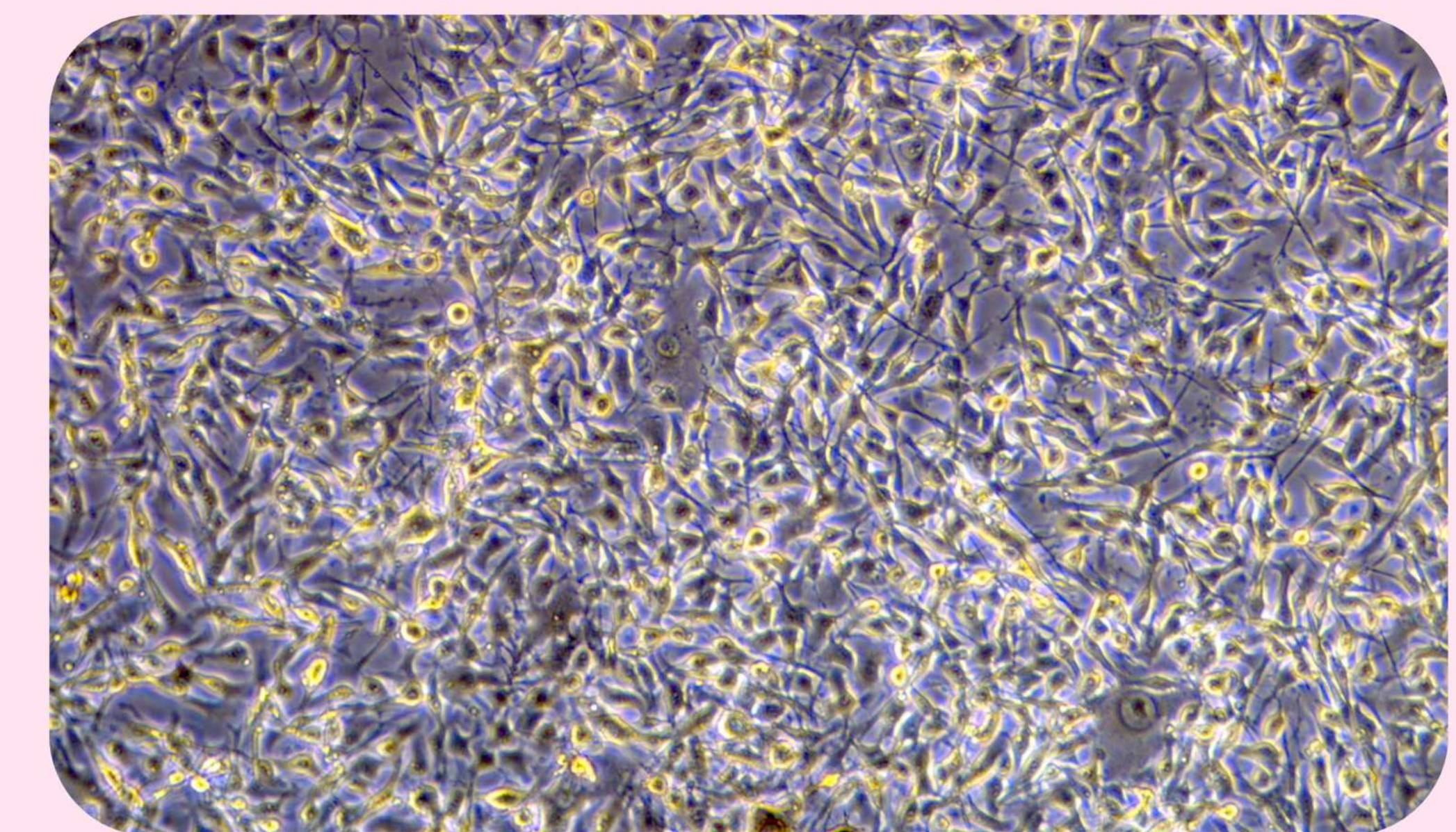


Figure 3. 10x Magnification Image of Cells with Melanin
Picture Taken by Student

References

- Culp, W. D., Tsagozis, P., Burgio, M., Russell, P., Pisa, P., & Garland, D. (2007). Interference of macrophage migration inhibitory factor expression in a mouse melanoma inhibits tumor establishment by up-regulating thrombospondin-1. *Molecular Cancer Research*, 5(12), 1225-1231. <https://doi.org/10.1158/1541-7786.MCR-07-0229>
- Maghfur, J., Olayinka, J., Hamzavi, I. H., & Mohammad, T. F. (2022). A Focused review on the pathophysiology of post-inflammatory hyperpigmentation. *Pigment Cell & Melanoma Research*, 35(3), 320-327. <https://doi.org/10.1111/pomr.13038>
- Sourmoy, L., Kindt, N., Ghanem, G., Saussez, S., & Journe, F. (2019). Role of Macrophage migration inhibitory factor (MIF) in Melanoma. *Cancers*, 11(4). <https://doi.org/10.3390/cancers11040529>